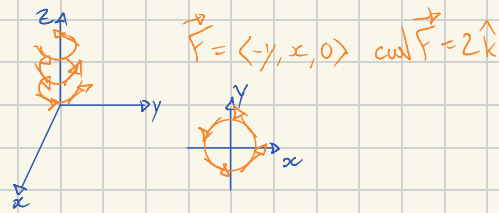
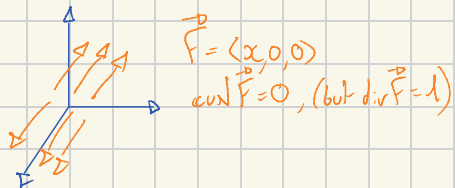
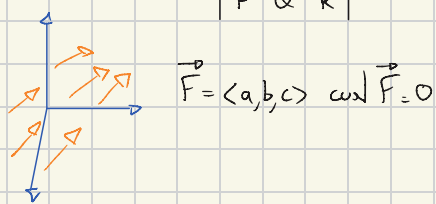


Lecture 31: Stokes' Theorem

Recall: $\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle$ (measures the rotation part of velocity field: direction = axis of rotation, magnitude = $2 \times$ angular velocity)



STOKES' THM:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$$

If C = closed curve

And $\vec{F}$ defined everywhere on S .

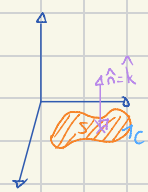
S = any surface bounded by C

Orientation: We need the orientations of C and S to be compatible: If: walk along C in the positive direction, with S to my left, then $\hat{n}$ is pointing upwards

OR: right-hand rule: Thumb along C positively: index tangent to S (towards interior of S); then, the middle finger will naturally point in the direction $\parallel$ to $\hat{n}$.

Comparing Stokes with Green:

S = portion of xy -plane, bounded by a curve C (counterclockwise)



$$\oint_C \vec{F} \cdot d\vec{r} = \oint_C P dx + Q dy + R dz \quad (z=0, dz=0) \stackrel{\text{Stokes}}{=} \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS = \iint_S (Q_x - P_y) dx dy$$

$\vec{F} = \langle P, Q, R \rangle$ $(\nabla \times \vec{F}) \cdot \hat{k} = z\text{-component of } \text{curl } \vec{F} = Q_x - P_y$

So in fact, Green's thm is just a special case of Stokes' thm in the xy -plane

Why Stokes is true?

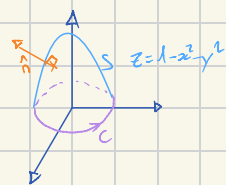
* we know it for C, S in xy -plane

* also for C, S in any plane (we are using the fact that work, flux, curl make sense independently of the coordinate system)

* given any S : decompose it into tiny, almost flat pieces: sum of work around each piece = work along C

Example: work of $\vec{F} = z\hat{i} + x\hat{j} + y\hat{k}$ around unit circle in xy -plane (counterclockwise) sum of flux through each piece = flux through S

Directly: $\oint_C z dx + x dy + y dz$ $x = \cos t$ $y = \sin t$ $z = 0 \Leftrightarrow \int_0^{2\pi} 0 + \cos t (\cos t dt) + 0 = \int_0^{2\pi} \cos^2 t dt = \pi$



$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS \quad \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & x & y \end{vmatrix} = \langle 1-0, 0+1, 1-0 \rangle = \langle 1, 1, 1 \rangle$$

$$\hat{n} dS = \langle -f_x, -f_y, 1 \rangle dx dy \Leftrightarrow \langle 2x, 2y, 1 \rangle dx dy$$

$$\iint_S \langle 1, 1, 1 \rangle \cdot \langle 2x, 2y, 1 \rangle dx dy = \iint_S (2x + 2y + 1) dx dy = (\text{switch to polar ...}) = \pi$$